

Brian R. Williams

Assistant Professor of Mathematics, Boston University

Department of Mathematics and Statistics

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Curriculum vitae | Updated October 2026

Research interests

Geometry and homotopical algebra in quantum field theory and string theory; factorization algebras, holomorphic field theories, supersymmetric twists, and higher-dimensional chiral and vertex algebras.

Academic appointments

Boston University Assistant Professor	2022–present
University of Edinburgh Whittaker Research Fellow	2020–2022
Mathematical Sciences Research Institute (now SLMath) Postdoctoral Research Fellow, Berkeley	2020
Northeastern University Zelevinsky Research Fellow	2018–2020

Education

Northwestern University Ph.D. in Mathematics Thesis: <i>The holomorphic σ-model and its symmetries.</i> Advisors: John Francis and Kevin Costello.	2012–2018
University of Florida B.S. in Mathematics and Physics, <i>summa cum laude</i>	2008–2012

Grants and fellowships

- **NSF Conference Grant DMS-2329854** (2023–present). New England Algebraic Topology and Mathematical Physics Seminar (NEAT MAPS).
- **NSF Graduate Research Fellowship** (2012–2018).

Visiting positions

- **Mittag-Leffler Institute**, visiting fellow, *Cohomological Aspects of Quantum Field Theory* (2025).
- **Ludwig-Maximilians-Universität München**, Center for Advanced Studies, visiting fellow (2024).
- **Perimeter Institute for Theoretical Physics**, fellow (2015).

Preprints and recent manuscripts

1. **Exceptional symmetry in six-dimensional superconformal quantum field theory**
With Ingmar Saberi.
2026. [PDF](#).
2. **Higher-dimensional Virasoro algebras**
With Zhengping Gui.
2026. [PDF](#).
3. **Two-dimensional Virasoro algebras**
With Zhengping Gui.
2026. [PDF](#).
4. **Higher-dimensional chiral algebras in the Jouanolou model and free-field realization**
With Zhengping Gui and Minghao Wang.
2025; revised 2026. [arXiv:2510.26608](#).
5. **Unravelling the holomorphic twist II: anomalies and extended supersymmetry**
With Pieter Bomans, Niklas Garner, and Jingxiang Wu.
2025. [arXiv:2509.16737](#).
6. **Local superconformal algebras**
With Fabian Hahner, Surya Raghavendran, and Ingmar Saberi.
2024. [arXiv:2410.08176](#).
7. **The local cohomology of vector fields**
Sole author.
2024. [arXiv:2405.05174](#).
8. **A complex geometric perspective on a, c anomalies**
Sole author.
2024. [arXiv:2404.16128](#).

Publications

1. **On the renormalization and quantization of topological-holomorphic field theories**
With Minghao Wang.
Advances in Mathematics **496** (2026), 110970. [DOI](#).
2. **Higgs and Coulomb branches from superconformal raviolo vertex algebras**
With Niklas Garner and Surya Raghavendran.
Advances in Mathematics **482**, Part A (2025), 110566. [DOI](#).
3. **Enhanced symmetries in minimally twisted three-dimensional supersymmetric theories**
With Niklas Garner and Surya Raghavendran.
Advances in Theoretical and Mathematical Physics **29** (2025), no. 4, 815–862. [DOI](#).
4. **Holomorphic field theories and higher algebra**
With Owen Gwilliam. Survey article.
Bulletin of the London Mathematical Society **57** (2025), no. 10, 2903–2974. [DOI](#).
5. **Raviolo vertex algebras**
With Niklas Garner.
Selecta Mathematica **31** (2025), no. 3, article 46. [DOI](#).
6. **Quantization of topological-holomorphic field theories: local aspects**
With Owen Gwilliam and Eugene Rabinovich.
Communications in Analysis and Geometry **32** (2024), no. 4, 1157–1231. [DOI](#).
7. **Semi-chiral operators in 4d $\mathcal{N} = 1$ gauge theories**
With Kasia Budzik, Davide Gaiotto, Justin Kulp, Jingxiang Wu, and Matthew Yu.
Journal of High Energy Physics (2024), no. 5, article 245. [DOI](#).
8. **Twisted eleven-dimensional supergravity**
With Surya Raghavendran and Ingmar Saberi.
Communications in Mathematical Physics **402** (2023), 1103–1166. [DOI](#).

9. **Batalin–Vilkovisky quantization and supersymmetric twists**
With Pavel Safronov.
Communications in Mathematical Physics **402** (2023), 35–77. [DOI](#).
10. **Superconformal algebras and holomorphic field theories**
With Ingmar Saberi.
Annales Henri Poincaré **24** (2023), no. 2, 541–604. [PDF](#).
11. **Constraints in the BV formalism: six-dimensional supersymmetry and its twists**
With Ingmar Saberi.
Advances in Mathematics **412** (2023), 108806. [PDF](#).
12. **Koszul duality in quantum field theory**
With Natalie Paquette.
Confluentes Mathematici **14** (2022), no. 2, 87–138. [PDF](#).
13. **Factorization algebras and abelian CS/WZW-type correspondences**
With Owen Gwilliam and Eugene Rabinovich.
Pure and Applied Mathematics Quarterly **18** (2022), no. 4, 1485–1553. [PDF](#).
14. **A taxonomy of twists of supersymmetric Yang–Mills theory**
With Chris Elliott and Pavel Safronov.
Selecta Mathematica **28** (2022), no. 4, article 73. [PDF](#).
15. **Perspectives on the pure spinor superfield formalism**
With Richard Eager, Fabian Hahner, and Ingmar Saberi.
Journal of Geometry and Physics **180** (2022), 104626. [PDF](#).
16. **Holomorphic Poisson field theories**
With Chris Elliott.
Higher Structures **5** (2021), no. 1, 282–309. [PDF](#).
17. **Higher Kac–Moody algebras and symmetries of holomorphic field theories**
With Owen Gwilliam.
Advances in Theoretical and Mathematical Physics **25** (2021), no. 1, 129–239. [PDF](#).
18. **Toroidal prefactorization algebras associated to holomorphic fibrations and a relationship to vertex algebras**
With Matt Szczesny and Jackson Walters.
Advances in Mathematics **386** (2021), 107799. [PDF](#).
19. **Chiral differential operators via quantization of the holomorphic sigma model**
With Vassily Gorbounov and Owen Gwilliam. Monograph.
Astérisque **419** (2020), ix + 210 pp. [PDF](#).
20. **Twisted characters and holomorphic symmetries**
With Ingmar Saberi.
Letters in Mathematical Physics **110** (2020), no. 10, 2779–2853. [PDF](#).
21. **Renormalization for holomorphic field theories**
Sole author.
Communications in Mathematical Physics **374** (2020), no. 3, 1693–1742. [PDF](#).
22. **The holomorphic bosonic string**
With Owen Gwilliam.
Contemporary Mathematics **718** (2018), 213–258. [arXiv:1711.05823](#).
23. **Homotopy RG flow and the non-linear σ -model**
With Ryan Grady.
Contemporary Mathematics **718** (2018), 187–211. [arXiv:1710.05973](#).
24. **Asymptotic freedom in the BV formalism**
With Chris Elliott and Philsang Yoo.
Journal of Geometry and Physics **123** (2018), 246–283. [PDF](#).
25. **The Virasoro vertex algebra and factorization algebras on Riemann surfaces**
Sole author.
Letters in Mathematical Physics **107** (2017), no. 12, 2189–2237. [PDF](#).

Doctoral advising and mentorship

Ph.D. advising, Boston University

- Garret Oren — co-advisor; graduation 2026.
- Tianchen Tang — expected graduation spring 2027.
- Cesai Li — expected graduation spring 2029.
- Yan Xuan — expected graduation spring 2029.
- Severin König — expected graduation spring 2030.

Ph.D. dissertation committees, Boston University

Greyson Potter (defended 2023); Sam Bardwell-Evans (defended 2023).

Undergraduate thesis advising

Boston University: Joseph Sorkin (2026), Jett Lackey (2026), Zoe Siegelnickel (2025), Zezheng Liu (2024).
University of Edinburgh: Lucy Spouncer (2022).

Student research community

Organizer, [student lunch seminar](#), Boston University.

Academic leadership and organization

- **NEAT MAPS conference series** (2023–2025). Co-organizer with Chris Elliott and Laura Murray. Semesterly meetings at Boston University, Providence College, Northeastern University, and Amherst College.
- **BU Geometry and Physics Seminar** (2022–present). Co-organizer with Siu-Cheong Lau, Boston University.
- **Joint AMS/UMI meeting, special session on Factorization Algebras and Geometry** (July 2024). Co-organizer with Chiara Damiolini, Zhengping Gui, and Matt Szczesny, Palermo, Italy.
- **Arithmetic Quantum Field Theory program** (January–March 2024). Co-organizer with David Ben-Zvi, Natalie Paquette, and Solomon Friedberg, Center of Mathematical Sciences and Applications, Harvard University.
- **Twisted Holography and Koszul Duality seminar** (fall 2021). Co-organizer with Dylan Butson and Surya Raghavendran.
- **Twisted QFT seminar** (2021). Co-organizer with Tudor Dimofte, University of Edinburgh.
- **Hodge seminar** (2020–2022). Organizer, University of Edinburgh.
- **Physical Mathematics of Quantum Field Theory** (2020). Co-organizer with Chris Elliott, Owen Gwilliam, and Matteo Lotito.
- **Joint MIT/Northeastern RTG seminar** (2018–2019). Co-organizer with Pavel Etingof and Valerio Toledano-Laredo. Topics: Yangians and gauge theory; factorization homology along surfaces.

Invited lecture series and advanced instruction

- **PROMYS**, Boston University (2025). Advanced lecture series on quivers and their representations for returning students and counselors.
- **Arithmetic Quantum Field Theory program**, CMSA, Harvard University (2024). *Algebraic quantum field theory*.
- **Arithmetic, Geometry, and Physics: Higher Algebraic Structures**, Korea Institute for Advanced Study (2022). *Twisted holography*.
- **LMS Autumn School** (2020). *Infinite-dimensional Lie algebras and their multivariable generalizations*.
- **Factorization Algebras and Functorial Field Theory**, Oberwolfach, Germany (2016). *Observables in the BV formalism*.
- **Physical Mathematics of QFT**, University of Massachusetts Amherst. Summer-school speaker for advanced graduate students.

Teaching

Boston University	2022–present
Differential geometry; differential topology; topics in geometry and physics; honors linear algebra; single-variable and multivariable calculus.	
Fall 2026: MA 563 and MA 225.	
University of Edinburgh	2020–2022
Course organizer for a topics course in complex geometry; tutor for Honours Algebra, Skills Lab, and Fundamentals of Pure Mathematics.	
Northeastern University	2018–2020
Linear algebra; topics in geometry and topology; Calculus I.	
Northwestern University	2014–2018
Honors linear algebra; complex variables; partial differential equations; calculus.	